



Microbial biosorption: a sustainable approach for metal removal and environmental remediation

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Abstract

The increasing prevalence of heavy metal pollution presents significant threats to ecosystems and public health, necessitating innovative and sustainable remediation strategies. Microbial biosorption has emerged as a promising and eco-friendly approach for removing and detoxifying heavy metals from contaminated environments. This review explores the unique biosorptive capabilities of microorganisms—specifically bacteria, fungi, and algae—highlighting their potential as cost-effective alternatives to conventional remediation methods that are often expensive and generate secondary pollutants. Advances in genetic engineering have further enhanced the biosorptive efficiency of these microorganisms, improving their utility in environmental remediation. Additionally, microbial biosorption offers the dual benefits of environmental detoxification and the recovery of valuable metals, contributing to the circular economy. This comprehensive analysis underscores the potential of microbial strategies in addressing heavy metal contamination and advocates for their integration into broader environmental remediation efforts. By providing a detailed overview of the mechanisms and advantages of microbial biosorption, this review aims to support the development of sustainable solutions to improve environmental health and ensure a sustainable future.

Keywords Biosorption · Environmental remediation · Heavy metals · Bacteria · Algae · Fungi

Introduction

Over the past 5 decades, human activities have severely disrupted the delicate equilibrium of Earth's natural systems—spanning the biosphere, lithosphere, hydrosphere, and atmosphere. Industrialisation, rapid urbanisation, and population growth have accelerated environmental degradation, culminating in a pervasive and insidious threat: heavy metal pollution (Joo et al. 2010; Medfu Tarekegn et al. 2020;

Zhou et al. 2023a; Jabbari et al. 2010). Unlike organic pollutants, heavy metals such as arsenic, cadmium, chromium, and lead do not degrade over time. Instead, they persist in the environment, accumulating in ecosystems and ultimately entering the food chain (Khan et al. 2015; Kermani et al. 2017; Zhou et al. 2023b). This persistence and its toxic and carcinogenic properties pose a severe risk to ecological integrity and public health (Fig. 1) (Fulekar et al. 2009; Hajdu et al. 2010; Namdeti 2023; Shokrzade and Saeedi Saravi 2009; Talos et al. 2009). Heavy metals' widespread soil, water, and air contamination is now recognised as a global crisis, demanding urgent and effective remediation strategies.

In addition to their environmental persistence, heavy metals are associated with a wide range of severe health consequences. While certain heavy metals have industrial applications, their hazards often outweigh their benefits. For instance, excessive chromium exposure has been linked to carcinogenic effects, and lead poisoning can cause cognitive impairments in children (Sun et al. 2015; Hou et al. 2015). Chronic mercury exposure may lead to Minamata disease, and prolonged cadmium intake is associated with itai-itai disease. Beyond these specific conditions, heavy metals can

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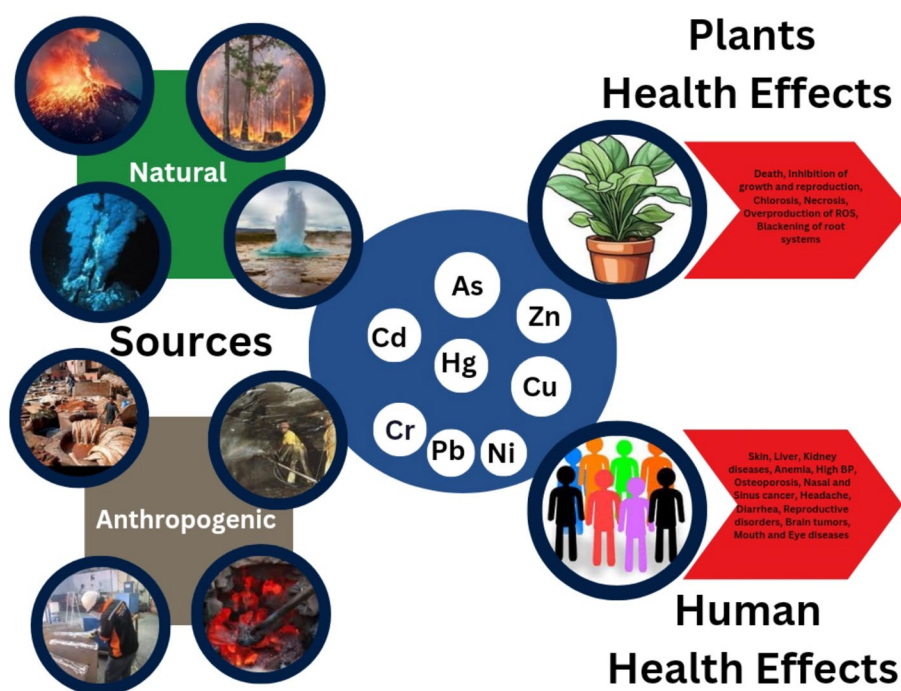
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Fig. 1 Natural and anthropogenic sources of heavy metals and their adverse effects on plant and human health



also compromise multiple human organ systems, causing nephrotoxicity, neurotoxicity, hepatotoxicity, skin damage, and cardiovascular issues. Proximity to industrial areas further heightens exposure risks, underscoring the need for preventive measures and sustainable remediation strategies (Mitra et al. 2022).

Traditional methods for heavy metal removal, including chemical precipitation, ion exchange, and reverse osmosis, have been widely employed to address this issue (Li et al. 2022; Namdeti 2023; Wołowiec et al. 2019). However, these techniques often come with significant drawbacks. High operational costs, energy-intensive processes, and generating secondary pollutants can limit their applicability, particularly in large-scale or resource-constrained settings. Additionally, these methods typically lack the specificity needed to target a wide range of metal ions, resulting in incomplete removal and further environmental hazards. In response, research has increasingly focused on identifying more sustainable and cost-effective remediation technologies, leading to the exploration of microbial biosorption as a promising alternative (Kumari et al. 2023; Sandil 2023; Sharif et al. 2022; Sharma et al. 2021a).

Biosorption, in particular, harnesses the natural capacity of microorganisms—such as bacteria, fungi, and algae—to sequester and detoxify heavy metals from contaminated environments. Compared to conventional decontamination methods, it offers significant advantages in terms of accessibility, efficiency, and cost-effectiveness (Koller and Saleh 2018). One of the key economic benefits of biosorption technology lies in the use of natural, readily available, and

affordable biomass, which often outperforms conventional decontamination methods in terms of efficiency and sustainability (Peter et al. 2018). Moreover, this environmentally friendly method does not generate harmful by-products while effectively targeting a broad spectrum of heavy metals (Li et al. 2022; Namdeti 2023; Sharif et al. 2022). Furthermore, microbial biosorption can be applied to various contaminated media, including water, soil, and industrial effluents, making it a versatile tool in the fight against pollution (Kumari et al. 2023; Sandil 2023; Sharif et al. 2022).

What sets this review apart from previous studies is its comprehensive analysis of the latest advancements in microbial biosorption, particularly in the context of genetic engineering and biotechnological innovations. While earlier reviews have focused on the general mechanisms of microbial metal uptake and the application of biosorption in environmental remediation (Tripathi et al. 2023; Taha et al. 2023; Ruiz-Fresneda et al. 2023; Ordóñez et al. 2023; Ojha et al. 2023; Nandini et al. 2023; Namdeti 2023; Gouda and Taha 2023; Chen et al. 2023; Yaashikaa et al. 2022; Tamjidi et al. 2022; Azeez et al. 2022), this review delves deeper into the recent breakthroughs that have significantly enhanced the biosorptive capacities of microorganisms. Advances in genetic engineering, for instance, have enabled the development of genetically modified strains with superior metal-binding properties, offering new avenues for the efficient remediation of heavily contaminated sites.

This review also emphasises the dual benefits of microbial biosorption: its role in environmental detoxification and its contribution to the circular economy (CE) through



the recovery and reuse of valuable metals. Biosorption is not only effective for removing even low concentrations of heavy metals but also for their recovery. For instance, a tool was developed to measure nutrient removal, carbon capture, and metabolite generation using *Chlorella vulgaris* and nitrifier-enriched-activated-sludge (NAS), showcasing the versatility of biosorption for different environmental applications (Sepehri et al. 2020). Furthermore, biosorption has been highlighted for its key advantages, including simple operation, no additional nutrient requirements, minimal sludge generation, lower operational costs, high efficiency, and the ability to regenerate biosorbent (Kanamarlapudi et al. 2018).

The principles of the CE further extend the potential of biosorption. CE emphasises the cascade use of biomass within systemic economic frameworks (Fig. 2), allowing bioproducts to transition from biowaste to reusable resources (Foundation 2013). Waste is increasingly recognised as a resource, particularly through its potential as a biosorbent for water treatment or metal recovery, aligning with CE principles (Foundation 2013). Moreover, integrating biowaste as a product to achieve zero-waste production presents a sustainable solution for modern industrial challenges. By leveraging the sorption potential of biowaste, biosorption can support the efficient treatment of waste, contributing to sustainability and advancing CE models (Madeła and Skuza 2021).

In addition to providing an updated overview of microbial biosorption mechanisms, this review critically assesses the environmental factors that influence biosorptive efficiencies, such as pH, temperature, and the presence of competing ions. By synthesising the most recent research findings, this work aims to inform future research directions and encourage the integration of microbial strategies into broader environmental remediation efforts. The review underscores the importance of a multidisciplinary approach, combining microbiology, environmental science, and engineering to develop sustainable solutions for heavy metal contamination.

As the global community grapples with the challenges posed by heavy metal pollution, the need for innovative and sustainable remediation strategies has never been more pressing. Microbial biosorption represents a cornerstone in these efforts, offering a green alternative to traditional methods and contributing to a cleaner, more sustainable future. This review aims to support the development of effective and sustainable solutions to improve environmental health and ensure sustainability for future generations by examining its mechanisms, advantages, and recent advancements.

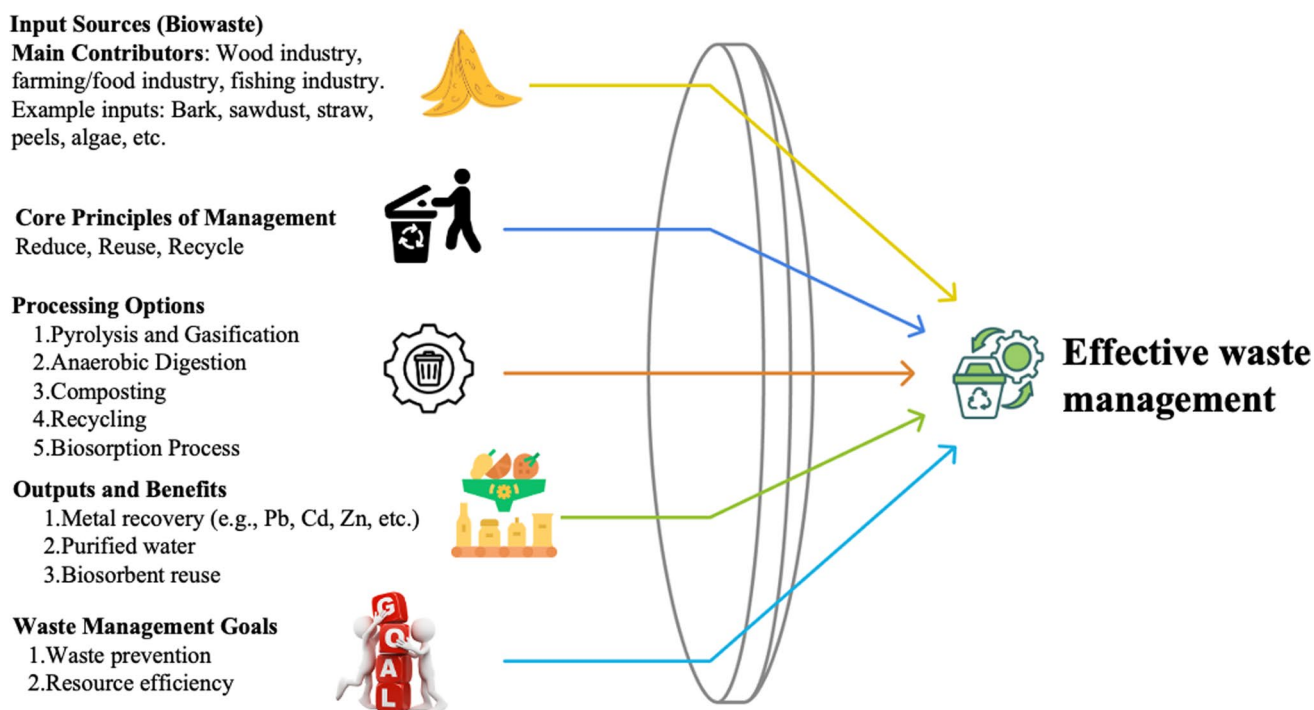


Fig. 2 Concept of managing biowaste towards a circular economy, illustrating key inputs, core management principles, processing options, outputs, and goals for effective waste management



Types of biosorbent microorganisms

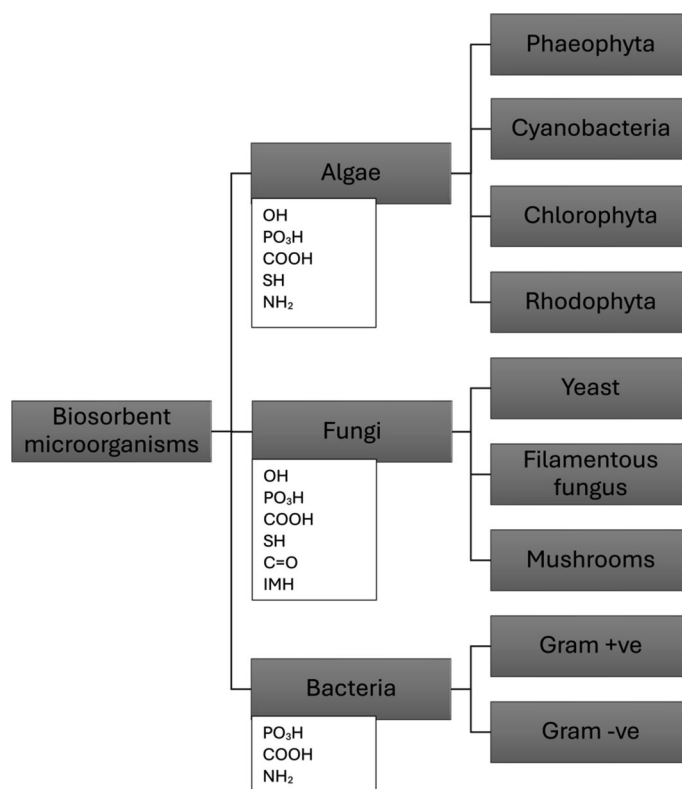
Efforts to mitigate the environmental impact of heavy metal pollution have led to the development of microbial biosorption strategies, leveraging the unique capabilities of algae, fungi, and bacteria to sequester and detoxify metals. These microorganisms are distinguished by their biological properties, widespread availability, and remarkable efficiency in binding heavy metals (Fig. 3). By classifying and understanding the mechanisms of microbial biosorption, this review sheds light on their critical role in the broader bioremediation landscape. Compared to chemical and physical separation methods, microbial biosorbents offer several advantages: they are biodegradable, renewable, cost-effective, adaptable to specific pollutants, and can often be deployed in situ, reducing the need for transportation and disposal of contaminated materials. This versatility enables microbial biosorption to mitigate contamination while protecting environmental and human health effectively.

Integrating microbial biosorption strategies into broader environmental remediation efforts involves complementing them with existing technologies, policies, and practices. For instance, microbial biosorption can remediate contaminated soils and groundwater or remove heavy metals and dyes from polluted environments, contributing

to ecological restoration. This technology has the potential to become a sustainable and cost-effective method for addressing contamination across various sources (Danouche et al. 2021). However, challenges remain, particularly the sensitivity of microbial systems to environmental changes. Factors such as temperature fluctuations, redox potential, moisture, and nutrient availability can disrupt microbial metabolism, reducing efficacy in bioremediation processes. Additionally, the long duration of microbial remediation efforts can hinder adoption due to economic constraints, as strategies with shorter time frames are often preferred to ensure a quicker return on investment (Wang et al. 2021). Balancing environmental and economic sustainability remains a critical consideration for practical applications. By understanding the diversity and capabilities of microbial biosorbents, researchers and practitioners can tailor bioremediation strategies to specific environmental challenges. By advancing scientific knowledge and technological innovation, this review highlights the importance of microbial diversity in maintaining ecological balance and driving sustainable solutions to environmental health challenges.

Bioremediation offers significant potential as a cost-effective and environmentally sustainable solution for addressing heavy metal contamination across diverse environments. Leveraging cutting-edge technologies such as genomic analysis, artificial intelligence, and precision monitoring

Fig. 3 Biosorbent microorganisms and the metal binding functional groups attached to their cell wall (SH: sulfhydryl, COOH: carboxyl group, OH: hydroxyl, NH₂: amino group, PO₃H: phosphate group, IMH: imidazole, C=O: carbonyl)



systems, bioremediation can enable precise and efficient degradation of pollutants in air, water, and soil (Agrawal et al. 2024). These advancements allow for tailored strategies that not only enhance remediation effectiveness but also optimise resource utilisation, reduce costs, and minimise environmental impact. To address current challenges and barriers, interdisciplinary and integrative collaboration remains essential for advancing bioremediation technologies. By merging scientific innovation with practical application, bioremediation emerges as a sustainable approach to restoring ecosystems, protecting public health, and mitigating the impacts of global climate change. This integrated approach underscores the urgency of adopting bioremediation strategies in addressing present and future environmental challenges (Mohan et al. 2024).

Algae

Algae, which are not true plants due to the absence of roots and stems, span a diverse range from microalgae to macroalgae. Their autotrophic nature allows them to thrive and produce substantial biomass even in conditions deficient in nutrients. Recognised for their significant biosorptive capabilities, algae stand out because of their large size, high sorption capacity, and the non-production of toxic substances, making them ideal for environmental remediation applications (Shamim 2018).

These organisms are systematically categorised into four primary groups based on the pigmentation of their cells: Chlorophyta (green), Phaeophyta (brown), Rhodophyta (red), and Cyanobacteria (blue-green). Further classification is based on various factors, including the structure of their cell wall, cellular localisation, and the characteristics of their flagella, highlighting the complexity and diversity of algal species (Ameri et al. 2020; Goswami et al. 2022; Ordóñez et al. 2023; Znad et al. 2022).

The cell wall composition of algae is critically important for their ability to sequester heavy metals from their environment. Green algae, as eukaryotic organisms, are noteworthy for their ability to use solar energy to transform atmospheric carbon dioxide into organic carbon. These predominantly freshwater-dwelling microalgae have cell walls mainly composed of cellulose, enriched with proteins intricately bound to polysaccharides, thus forming glycoproteins (Ameri et al. 2020; Miazek et al. 2015; Romera et al. 2007).

Brown algae, which may exist as single cells or in colonies displaying yellow or golden-brown hues, are characterised by their unique motility facilitated by two dissimilar flagella. Predominantly marine organisms, these macroalgae are known for their complex, macroscopic seaweeds, with a brown colouration derived from carotenoid pigments and fucoxanthin. Their cell walls comprise cellulose, alginic acid, and guluronic acids, predominantly featuring carboxyl

and sulphated groups, which are implicated in their superior metal uptake capabilities compared to other algal classes (Ameri et al. 2020; Ordóñez et al. 2023). Red algae, known for their resilience in thermal vents and for containing cellulose and sulphated polysaccharides like agar and carrageenans, represent another distinct class within algal taxonomy. Meanwhile, blue-green algae, which can appear as single cells, in batches, or as filamentous formations, display various colours, including light blue, olive green, and brown, and are characterised by their sporadic movements (Ameri et al. 2020; Ordóñez et al. 2023; Raize et al. 2004).

Among these classes, brown algae are particularly recognised for their enhanced capacity to absorb heavy metals. The efficacy of metal ion binding to algal surfaces is influenced by factors such as the specific algal species, the ionic charge of the metal, and the chemical makeup of the metal solution. Key binding sites on algal surfaces include functional groups such as sulfhydryl, hydroxyl, phosphate, sulphate, imidazole, amine, and carboxyl (Fig. 3), underscoring the biochemical diversity and potential of algae in bioremediation efforts (Shamim 2018).

In biosorption, algal blooms are formidable agents in wastewater treatment, particularly adept at mitigating or removing toxic heavy metals. This capability is crucial, given the significant environmental challenge posed by the accumulation of heavy metals in aquatic ecosystems, which can detrimentally affect living organisms due to their toxicity, carcinogenicity at even minimal concentrations, and propensity for bioaccumulation stemming from non-biodegradable nature (Ordóñez et al. 2023). Anthropogenic activities, including mining, smelting, and various manufacturing processes, are primary contributors to the escalation of heavy metal contamination in soils and groundwater (Arumugam et al. 2018).

Cadmium, highlighted for its frequent appearance in related publications—constituting about 23% of reports—underscores the pressing need for efficient removal strategies due to its environmental toxicity (Ordóñez et al. 2023). Copper and lead are closely followed, implicated in 20% and 19% of studies, respectively, highlighting their prevalence in industrial discharges. Conversely, metals like iron, receiving lesser attention at 3%, suggest a reliance on cost-effective precipitation methods for their removal, thus reducing the focus on biosorption exploration for such metals.

The study of brown algae garners attention for their inherent metal-affinity properties. Yet, recent research trends are shifting towards exploring green and red algae, indicating an expanding interest in diversifying biosorption research sources. *Cladophora* macroalga, representing green algae, has been studied for its efficacy in removing Cu(II), Cr(III), and Cd(II) from aqueous solutions. Investigations into pH levels, contact time, and mixing speed—keeping a constant temperature of 20 °C—revealed that an adsorbent dose of



1 g/L and initial metal ion concentrations set at 10 mg/L are optimal. Specifically, pH values of 7.0 for Cu(II), 5.0 for Cr(III), and 5.2 for Cd(II) led to removal efficiencies of 70, 80, and 85%, respectively, at a 120-min contact time (Bahaa et al. 2019). In another study, *Chlorella coloniales* was utilised to extract Cr, Cd, Co, Fe, and As ions from water, examining metal ion concentrations (5.0–15 mg/L), contact times (3–120 h), and adsorbent doses (1.0–3.0 g/L) via the Box–Behnken method. The findings indicated optimal contact times of 109, 95.6, 102, 111, and 108 h, and optimal adsorbent doses of 2.70, 2.71, 2.46, 2.72, and 2.91 g/L, resulted in removal efficiency of 96.8%, 98.7%, 98.5%, 95.4%, 97.3%, and 98.0% for the removal of Cr, Cd, Co, Fe, and As, respectively (Jaafari and Yaghmaeian 2019).

Chlorella vulgaris responds to cadmium and zinc exposure by inducing metallothionein (MT)-like proteins, crucial for bioaccumulating such heavy metals. This induction intensifies with greater metal stress, as analysed through high-performance liquid chromatography (Huang et al. 2009). The growth modulation of microalgae, requiring temperatures of 20–30 °C, by various metals, including iron and zinc, further illustrates the intricate relationship between algal growth and metal presence (Miazek et al. 2015). Moreover, a study demonstrated the reduction of Cr(VI) in *Chlorella vulgaris* through chromium reductase and glutathione (GSH), with overexpression of enzymes such as peroxidases (Px), superoxide dismutase (SOD), catalase (CAT), and glutathione reductase in response to metal exposure, underscoring the physiological adaptations of algae to detoxify their environment (Vingiani et al. 2019).

A different study revealed that brassinolide, a phytohormone, significantly mitigates the adverse effects of heavy metal stress on *Chlorella vulgaris* (Bajguz 2010). It promotes cellular growth through the enhanced activity of antioxidant enzymes like CAT and SOD, pointing to the potential of such compounds in enhancing algae's resistance to heavy metal toxicity.

Through these studies, algae's role in biosorption is enlightened, showcasing their potential in wastewater treatment and environmental remediation, with specific conditions and adaptations critical to maximising their biosorptive capabilities against a backdrop of heavy metal pollution. The biosorption capacity of some other species of green algae is provided in Table 1.

Heavy metals, including Cr, Ni, Cu, As, Cd, Hg, and Pb, present significant environmental hazards worldwide, with Pb and Cr notably prevalent in wastewater, posing substantial toxicity concerns (Ordóñez et al. 2023). Investigating the biosorption capabilities of brown seaweed species, *Hydroclathrus clathratus* and *Cystoseira barbata*, it was discovered an inverse correlation between metal concentration and biosorption capacity, with maximum uptake values for Cr and Pb recorded at 7.19 and 4.97 mg/g for *H. clathratus*,

and 7.30 and 4.61 mg/g for *C. barbata*, respectively (Ali et al. 2020).

Macrocystis pyrifera was used for the removal of Zn and Cd from solutions, demonstrating uptake capacities that align with those observed in other studies involving *Sargassum filipendula*, *Gymnogongrus torulosus*, and *Fucus vesiculosus* (Plaza Cazón et al. 2012). Notably, *M. pyrifera* and *Undaria pinnatifida* also efficiently removed Cr and Hg, highlighting the role of carboxylic and amino groups in Cr binding, whereas amino and sulfhydryl groups were key in Hg adsorption. In a related application, *Sargassum* sp. was shown to possess substantial biosorption capacities for Pb and Cu from stormwater, achieving capacities of 196.1 and 84.0 mg/g, respectively, with infrared spectroscopy analysis indicating a significant contribution of carboxyl groups to the biosorption process (Perumal et al. 2007).

The use of the brown alga *Padina sanctae-crucis* in removing Cu(II) and Co(II) ions from aqueous solutions was evaluated in a range of conditions, including pH levels (2–11), contact time (5–200 min), temperature (30–50 °C), and adsorbent dose (1–7 g/L), with an initial metal ion concentration of 20 mg/L and a mixing speed of 200 rpm (Foroutan et al. 2018). The optimal conditions were identified at a pH of 6, a contact time of 80 min, a temperature of 50 °C, and an adsorbent dose of 4 g/L, resulting in adsorption efficiencies of 96.68% for Cu(II) and 94.56% for Co(II) ions. This study affirms the potential of *Padina sanctae-crucis* as an effective biosorbent for Cu(II) and Co(II) removal from water, underscoring the value of brown algae in environmental remediation efforts. Additional insights into the biosorption capacity of brown algae are detailed in Table 1.

Cadmium removal from aqueous solutions has been effectively achieved using the red alga *Ceramium virgatum*. This alga exhibited a maximum biosorption capacity for Cd(II) ions of 39.7 mg/g, highlighting its potential in heavy metal remediation (Sari and Tuzen 2009). In a separate investigation, the biosorption capabilities of the treated red algae *Gracilaria fisheri* towards cadmium and copper was assessed, observing maximal absorption rates of 0.63 and 0.72 mmol/g, respectively (Chaisuksant 2003). Notably, the biosorption efficiency increased with pH, stabilising at pH 4. Additionally, the biosorption process for both metals was rapid, with 90% of the biosorption occurring within 30 min of exposure.

In contrast, another study focused on the utilisation of alginate extracted from the marine red algae *Callithamnion corymbosum* for the effective removal of Cu(II), Co(II), and Zn(II) ions from aqueous solutions (Bulgariu and Lucaci 2018). The reported maximum biosorption capacities for these metals were ranked as follows: Cu(II) at 64.52 mg/g, Zn(II) at 37.04 mg/g, and Co(II) at 18.79 mg/g. The optimal biosorption conditions were identified at pH 4.4, with an adsorbent dosage of 2.0 g/L, and conducted at ambient



Table 1 Biosorption of heavy metals by *Chlorophyta*, *Phaeophyta*, *Rhodophyta* and *Cyanobacteria*

Type of algae	Algal species	Heavy metal	Functional group/ enzymes	Optimal conditions	Efficiency	Refs.
Chlorophyta	<i>Codium vermilara</i>	Cu	–	Algal dose 0.75 g/L, pH 5.3, contact time 70.51 min	85.5%	Fawzy (2020)
	<i>Chlorella vulgaris</i>	Cu	Amine and carboxyl	pH 7, contact time 105 min., concentration 20 ppm	97.59%	Indhumathi et al. (2018)
	<i>Cladophora glomerata</i>	Cr	–	pH 5, biosorbent dose 0.1 g/L, contact time 200 min	100%	Godlewska et al. (2018)
	<i>Chlorella miniate</i>	Cr	Glutathione synthase/ SODs	pH 2.0, biomass 5.0 g/L, contact time 2 h	65%	Han et al. (2007)
	<i>Spirogyra</i> sp	Cr	–	pH 5.0, biomass dose of 2.5 g/L, contact time 45 min	81%	Bishnoi et al. (2007)
		Pb	Phytochelatin synthase/ MT	pH 5.0, 100 min, algal dose 0.5 g/L	Not defined	Gupta and Rastogi (2008)
	<i>Strichoccus bacillaris</i>	As	Phytochelatin synthase /SH	pH 6.8, contact time 24 h	Not defined	Pawlik-Skowrońska et al. (2004)
	<i>Scenedesmus armatus</i>	Cd	Hydroxyl and amine	pH 5, contact time 60 min	83%	Waqar et al. (2023)
	<i>Scenedesmus vacuolatus</i>	Cd	Phytochelatin/glu-tathione	–	Not defined	Le Faucheur et al. (2005)
	<i>Chlorella</i> sp.	Pb	Hydroxyl, carboxyl, amine and phosphate	pH 6, contact time 180 min, dose 1.5 g/L	78%	Molazadeh et al. (2015)
	<i>Ulva lactuca</i> sp	Cd	Glutathione Synthase	pH 5, contact time 60 min, dose 0.1 g/L	29.2 mg/g	Bulgariu et al. (2015)
		Pb	Phytochelatin synthase	pH 5, contact time 60 min, dose 0.1 g/L	34.7 mg/g	
Phaeophyta	<i>Sargassum muticum</i>	Zn	Phytochelatin synthase/ MT	pH 5, contact time 60 min, dose 1 g/L	32.0 mg/ g	Freitas et al. (2008)
	<i>Chaetoceros</i> sp.	Pb	Hydroxyl, carboxyl, amide and phosphate	pH 6, contact time 180 min, dose 1.5 g/L	60%	Molazadeh et al. (2015)
	<i>Lessonia nigrescens</i>	Cu	Carboxyl, hydroxyl, sulfonate and amide	pH 5, contact time 120 min	60.4 mg/g	Cid et al. (2018)
	<i>Phaeodactylum tricornutum</i>	Cd	Phytochelatin/glu-tathione	pH 8.1	Not defined	Kawakami et al. (2006), Morelli and Scarano (2001)
		Cu	Phytochelatin	–	Not defined	Morelli and Scarano (2001)
		Pb	Phytochelatin	–	Not defined	
	<i>Sargassum</i> sp.	Cd	SOD	pH 5	84.4 mg/g	Hajar (2009)
		Ni	–	pH 4.5	Not defined	Barquilha et al. (2019)
		Cu	–			
	<i>Padina pavonica</i>	Al	–	pH 4.5, contact time 60 min, biomass 8 g/L	77.3 mg/g	Sari and Tuzen (2009)
Rhodophyta	<i>Kappaphycus alvarezii</i>	Cr	Hydroxyl, carboxylic, sulfoxide, amino, aromatic	pH 7, contact time 70–80 min	62.93%	Pandya et al. (2017)
		Ni		–	90.22%	
		Cu		–	81.53%	



Table 1 (continued)

Type of algae	Algal species	Heavy metal	Functional group/ enzymes	Optimal conditions	Efficiency	Refs.
	<i>Gracilaria changii</i>	Pb	Hydroxyl, carboxylic, sulfoxide	pH 4.5, contact time 115 min, biomass 1 g/L	96.3%	Isam et al. (2019)
		Cu		pH 5, contact time 45 min, biomass 1 g/L	44.8%	
Cyanobacteria	<i>Spirulina platensis</i>	Cu	SOD/MT	pH 6.7, biomass 2 g/L	91%	Solisio et al. (2006)

temperature, underscoring the specificity of biosorption parameters for achieving maximum efficacy.

The significant potential of the cyanobacterium *Anabaena* sp. for the biosorption of Cd(II) and Pb(II) ions from aqueous solutions was investigated (Adeli et al. 2017). It was identified that the most effective conditions were pH levels of 3 and 5.5, contact times of 90 and 60 min, a consistent ambient temperature of 25 °C, and adsorbent doses of 0.2 and 0.1 g/L. Under these parameters, *Anabaena* sp. demonstrated remarkable efficiencies, removing 94.3% of Cd(II) and 88.6% of Pb(II) ions at a heavy metal concentration of 50 mg/L, showcasing its potential as an efficient agent in heavy metal remediation efforts.

The biosorption capabilities of another cyanobacterium, *Anabaena sphaerica*, to sequester Cd(II) and Pb(II) ions from aqueous environments were also explored (Abdel-Aty et al. 2013), with capacities reaching 111.1 mg/g for Cd(II) and 121.95 mg/g for Pb(II), respectively. A detailed examination using infrared spectroscopy further elucidated the mechanism behind this biosorption, revealing that the amino, carboxyl, hydroxyl, and carbonyl functional groups on the surface of *Anabaena sphaerica* play pivotal roles in binding these metal ions. This analysis confirms the alga's efficacy in removing hazardous ions from water. It provides insight into the biochemical interactions facilitating this process, contributing to a deeper understanding of cyanobacteria's role in environmental decontamination strategies.

In the complex interplay between microalgae and heavy metals, phytochelatin emerges as crucial biomolecules, functioning as cysteine-rich peptides akin to Class II MTs—or Class III in certain delineations—synthesised as polypeptides. Small yet potent metal-binding peptides are synthesised via phytochelatin synthase, playing an integral role in the sequestration of heavy metals (Goswami et al. 2022; Miazek et al. 2015). The remediation process also involves a variety of specific functional groups such as sulphate, phosphate, amino, amide, thioether, amine, carbonyl, phenol carboxyl, hydroxyl, and sulfhydryl, with the carboxyl group noted for its critical importance in metal chelation (Spain et al. 2021). An illustrative example of

functional group intervention is the addition of ethylenediaminetetraacetic acid at a concentration of 34 µM to cultures of *Scenedesmus* sp., mitigating growth inhibition caused by 40 µM Cu. However, Cd exposure without ethylenediaminetetraacetic acid significantly restricts algal growth (Miazek et al. 2015).

Incorporating organic compounds such as phytohormones and chelating agents like phytochelatin into algal cultures promotes growth and enhances metal accumulation capabilities, enabling microalgae to resist and flourish amidst heavy metal exposure. A seminal study delved into the dynamics of thiols—gamma-glutamylcysteine, GSH, and phytochelatin—in *Scenedesmus vacuolatus* under metal and metalloid stress, including Cu, Zn, Ni, Pb, Ag, As(V), Sb(III), and Sb(V) (Le Faucheur et al. 2006). Glutathione, a pivotal thiol responsible for enzymatic regulation within algal cells, showed increased levels upon exposure to Pb, Ag, As(III), and As(V), underscoring its essential role in cellular defence mechanisms (Fahey and Newton 1987). Phytochelatin induction was notably high when *Scenedesmus* cells were subjected to Cu, Zn, Ni, and Pb. Intriguingly, the activity of phytochelatin synthase—key for converting GSH into phytochelatin—was augmented in the presence of Zn, Cu, Ag, Cd, Pb, As, and Sb. However, Ni exposure did not stimulate enzyme production, highlighting the specificity of enzyme-metal interactions (Ahner et al. 1995; Ahner and Morel 1995).

The metal type and the reacting thiol in its synthesis influenced the enzyme's activation intensity. Notably, metalloids such as Sb(III) and As(V) induced phytochelatin production, suggesting *Scenedesmus*'s capability for metalloid detoxification. Exposure to As(V) led to the generation of GSH. However, an inverse relationship was observed between As(V) concentration and phytochelatin levels, indicating that low As(V) concentrations (less than 8 µM) stimulate phytochelatin production. In contrast, higher concentrations inhibit it, shedding light on the delicate balance of thiol-mediated detoxification pathways in microalgae (Ahner et al. 1995; Ahner and Morel 1995).



Fungi

Fungi, a taxonomically diverse group of eukaryotic organisms ranging from unicellular yeasts to complex filamentous forms characterised by branching hyphae, exhibit varying degrees of biosorption efficacy. Given that sorption is primarily a surface-mediated process, factors such as the ionic state of the biomass, surface area, and polarity of the biosorbent play critical roles in determining the effectiveness of fungal biosorption (Dhankhar and Hooda 2011).

Fungi, encompassing yeasts, mushrooms, moulds, and other eukaryotic organisms, are endowed with cell wall structures that possess significant metal-binding properties, making both living and non-living fungal forms potent biosorbents for metal sequestration (Namdeti 2023; Sulaymon 2014; Wang and Chen 2006). Metal uptake in fungi occurs through two primary pathways: (i) bioaccumulation, or active uptake, which is metabolically driven and exclusive to living cells, and (2) biosorption, or passive uptake, characterised by the adsorption of metal ions onto the cell wall surface, independent of cellular metabolism. This distinction underscores the complex interplay between fungal bioaccumulation and biosorption, which can be influenced by environmental conditions such as temperature and the presence of metabolic inhibitors (Namdeti 2023; Shamim 2018; Bellion et al. 2006).

The biosorption process may engage metal ions through interactions with cell surface functional groups, including ion exchange, complexation, and physical adsorption. Moreover, fungi employ both extracellular and intracellular methods for metal uptake: extracellularly, metals are bound to cell walls or precipitated through chelation, whereas intracellularly, metals are bound to compounds such as organic acids, peptides, phosphates, and sulphurs before being transported into the cell. This strategy effectively reduces metal toxicity, enhancing the microorganisms' defence mechanisms and facilitating the accumulation of metals (Bellion et al. 2006).

Fungi's remarkable bioremediation capabilities extend beyond metal absorption, encompassing the secretion of various enzymes. These extremophilic enzymes enable fungi to thrive in varied and often harsh environmental conditions, from soil to marine and freshwater habitats. This further highlights their ecological and biotechnological significance in metal sequestration and environmental restoration efforts (Deshmukh et al. 2016).

In exploring fungal biosorption capabilities, the composition of the cell wall is identified as a pivotal factor due to its exceptional metal-binding attributes. The fungal cell wall is intricately composed of chitins, mannans, and glucans, alongside many lipids, polysaccharides, and pigments such as melanin, significantly augmenting its biosorption capacity (Das et al. 2008; Shamim 2018). Predominantly, the cell wall comprises polysaccharides, representing up to

90% of its structure. Functional groups, including carboxyl, phosphate, uranic acids, proteins, and nitrogen-containing ligands, notably chitin or chitosan, are instrumental in the binding of metal ions (Jamir et al. 2024; Gouda and Taha 2023; Namdeti 2023; Sulaymon 2014).

The biosorptive potential of fungal cells can be further augmented or altered through a variety of physical or chemical treatments, such as autoclaving, heating, and the use of agents like dimethyl sulfoxide, laundry detergent, orthophosphoric acid, formaldehyde, glutaraldehyde, and sodium hydroxide (Das et al. 2008). Within the fungal domain, macrofungi, often called mushrooms, demonstrate a significant metal uptake capacity. These organisms, thriving in pristine forests and pollutant-laden environments, adeptly sequester metals within their fruiting bodies, mycelia, and sporocarps, showcasing their environmental adaptability (Namdeti 2023; Lu et al. 2020; Shamim 2018).

Chitin, an essential polysaccharide in the fungal cell wall composed of *N*-acetylglucosamine residues, exemplifies the interaction dynamics within the fungal cell wall. The positive charge of chitin or chitosan engaging with negatively charged phosphates and glucuronic residues underscores the strategic exploration into fungal biosorption mechanisms for heavy metal sequestration (Dusengemungu et al. 2020). Notably, yeast and filamentous fungi emerge as the primary fungal groups leveraged for the biosorption of heavy metals, highlighting their potential in bioremediation strategies.

Yeast, particularly *Ascomycota* phylum, is an invaluable unicellular fungal model for exploring the physiological and molecular mechanisms underlying eukaryotic cell interactions with heavy metals. For instance, *Saccharomyces cerevisiae* has demonstrated biosorption capacities of 40–80 mg/g for Pb, 10–60 mg/g for Cd, and 5.5 mg/g for Cr ions. Similarly, *Candida albicans* exhibits efficient biosorption of multiple heavy metals, with 76% for Cr, 57% for Pb, and 46% for Cd (Jamir et al. 2024). These examples underscore the practical applications of yeast in remediating heavy metal-contaminated environments. Yeast exhibits a superior biosorbent capacity for removing heavy metal ions compared to bacteria, filamentous fungi, and microalgae (Bahafid et al. 2017). This elevated capacity is primarily attributed to yeast's rapid proliferation and the structural peculiarities of its cell wall. The biosorption efficiency in yeast is facilitated mainly by electrostatic interactions, thanks to its negatively charged cell surface abundant in phosphates and carboxyl groups, which avidly bind metal ions (Jamir et al. 2024).

Several yeast species have been recognised for their proficiency in metal sequestration, affirming their role in the bioremediation of metals such as Zn, Pb, Cd, Co, Cr, Ag, and U. Specifically, genera such as *Saccharomyces*, *Candida*, and *Pichia* are distinguished for their biosorption capabilities (García-Béjar et al. 2020). *Saccharomyces cerevisiae*, in particular, is acclaimed as a paradigmatic system



for biosorption studies due to its non-pathogenic nature, ease of cultivation, high biomass yield on simple media, and the availability of its complete genome sequence. These attributes facilitate genetic manipulations and render it a prime model for molecular biology investigations (Jamir et al. 2024; Gouda and Taha 2023; Namdeti 2023).

The biosorption performance of yeast is modulated by multiple factors, including the physicochemical attributes of the metal ions (e.g., valency, ionic radius), yeast cell age, and the specifics of the culture conditions (e.g., the composition of the growth medium and carbon source). Operational parameters such as initial metal concentration, biomass concentration, temperature, pH, and the presence of competing ions in the medium also critically influence biosorption efficiency (Tamjidi et al. 2022; Shamim 2018). Furthermore, the comparative size advantage of yeast cells over other microorganisms enhances their applicability in metal bioremediation efforts. Various configurations of *S. cerevisiae*, including immobilised vs. free cells, living vs. dead cells, and genetically engineered vs. non-engineered cells, have been scrutinised for their biosorptive properties, highlighting the flexibility and potential of yeast in heavy metal remediation processes (Shamim 2018; Jamir et al. 2024; Namdeti 2023; Gouda and Taha 2023). The contributions of different yeast types to heavy metal removal are detailed in Table 2.

Filamentous fungi, distinguished by their development into tubular hyphae, emerge as macroscopic entities exhibiting a broad array of morphological extensions across different phyla (Jamir et al. 2024). These organisms are celebrated for their multifaceted roles as biofactories in producing enzymes, antibiotics, and sustainable materials that present viable alternatives to conventional plastics. Notably, their intrinsic capacity to decompose organic compounds and synthesise secondary metabolites positions filamentous fungi, including *Neurospora crassa*, at the forefront of research for applications ranging from antimicrobial development to environmental remediation (Wösten 2019).

The employment of filamentous fungi in industrial settings offers substantial benefits, including cost-efficiency, resource abundance, and minimal environmental impact. The expansive mycelial networks of these fungi significantly augment their biomass and surface area, enhancing their utility in a spectrum of biotechnological processes (Dusengemungu et al. 2020). *Neurospora crassa*, in particular, has been scrutinised for its biosorptive capabilities, illustrating that metal sequestration involves cell wall binding and protein interactions, facilitating metal accumulation and efflux (Latha et al. 2012). *Aspergillus* species, capable of tolerating heavy metal concentrations up to 1500 mg/L, underscore the robustness of filamentous fungi in environmental detoxification. *Aspergillus foetidus*, for instance, has demonstrated the capacity to absorb significant quantities

of Pb (approximately 200 mg/L), leveraging its tolerance to mitigate wastewater toxicity (Deshmukh et al. 2016).

Filamentous fungi such as *Aspergillus spp.*, *Trichoderma spp.*, *Rhizopus spp.*, and *Penicillium spp.* have been extensively investigated, revealing unique attributes and potential industrial applications (Table 3). The fungal cell wall surface, replete with metal-attracting functional groups including carboxyl, amine, hydroxyl, phosphate, and sulfhydryl, has spurred interest in utilising fungal biomass as a biosorbent. This attraction is driven by the diverse absorptive mechanisms these functional groups facilitate for heavy metal remediation (Gouda and Taha 2023; Legorreta-Castañeda et al. 2022). Fungi exhibit resilience against environmental fluctuations, such as changes in pH, aeration, and temperature, while maintaining their metal-binding efficiency, distinguishing them from other microorganisms, mainly bacteria (Dhankhar and Hooda 2011; Sri Lakshmi Ramya Krishna et al. 2018).

Furthermore, the efficacy of fungi as metal sorbents and the valorisation of industrial fermentation waste as fungal biomass underscore the feasibility of scaling fungal biosorption processes. This scalability is particularly pertinent for the remediation of metal-contaminated effluents, leveraging cost-effective growth media (Leitão 2009; Dusengemungu et al. 2020; Taha et al. 2023). It was highlighted that most fungal biosorbents are characterised by their non-toxicity, affirming their safety, eco-friendliness, and suitability for diverse applications (Sri Lakshmi Ramya Krishna et al. 2018). Challenges remain in identifying optimal fungal species for in-situ applications and bridging the gap in our understanding of biosorption mechanisms. The transition from controlled laboratory settings to real-world environmental conditions necessitates further exploration to fully harness the potential of fungal biosorption on a large scale (Jamir et al. 2024).

Bacteria

Bacteria, ubiquitous unicellular microorganisms, often engage in symbiotic relationships with other organisms. Their structural simplicity, marked by the absence of a true nucleus and membrane-bound organelles, contrasts with the complexity of their cell wall composition—a key determinant of cellular integrity. This composition forms the basis for the binary classification of bacteria: Gram-positive bacteria, with a substantial peptidoglycan layer (20–80 nm) connected by amino acid bridges, and Gram-negative bacteria, which have a thinner peptidoglycan layer (10–15 nm) and less extensive cross-linking (Priyadarshane and Das 2021). The cell wall of Gram-negative bacteria is negatively charged, owing to the presence of lipopolysaccharides, teichoic acids, and teichuronic acids. Not all Gram-negative bacteria possess extracellular polysaccharides with



Table 2 Biosorption of heavy metals by different types of yeast

Microbial species	Heavy metal	Mechanism	Optimal conditions	Efficiency	Refs.
<i>Rhodotorula mucilaginosa</i>	Ag(I), Ag(CN) ₂ ⁻	Biosorption via carbonyl groups, glucose-dependent uptake for Ag(CN) ₂ ⁻	MES buffer (chloride-free), pH 6.5, Temperature 25 °C, Glucose 2% for Ag(CN) ₂ ⁻ uptake	Maximum uptake: Ag ⁺ (22.20 mg/g for UFMG-Y27), Ag(CN) ₂ ⁻ (glucose-dependent). Carbonyl groups identified as binding sites	Gomes et al. (2002)
<i>Pichia pastoris</i>	Cu(II)	Biosorption via cell wall, membrane, and cytoplasm components	pH: 6.0, Temperature: 25 °C, Cu ²⁺ solution: 100 mg/L, Equilibrium time: 15 min	Maximum removal efficiency: 41.1%, Biosorption capacity: 6.2 mg/g. Cell wall: 21.2%, membrane: 20.7%, cytoplasm: 18.5%	Chen et al. (2021)
	Zn, Ni	Chelation by cell wall components and intracellular accumulation	pH 5–6, 25 °C, 24 h contact time	Zn: ~70%, Ni: ~65%	Wang and Chen (2009)
<i>Pichia guilliermondii</i>	Cr(III), Cr(VI)	Biosorption, riboflavin synthesis enhances Cr resistance	Medium: Modified YPD, pH 5.5, Temperature: 30 °C, Cr concentrations: Cr(III) 2–9 mM, Cr(VI) 0.3–0.5 mM	Cr(III): Uptake up to 10 mg/g dry weight, IC30 up to 9 mM. Cr(VI): Uptake up to 0.9 mg/g dry weight, IC30 up to 0.5 mM. Riboflavin production: Up to 10.9 mg/g dry weight under Cr(VI)	Ksheminska et al. (2003)
<i>Candida sphærica UCPO995</i>	Pb, Zn, Fe	Biosurfactant-mediated chelation and desorption	0.25% biosurfactant, 27 °C, pH 7, 24 h contact time	Pb: 79%, Zn: 90%, Fe: 95%	Luna et al. (2016)
<i>Candida lipolytica</i>	Cu	Bioaccumulation via ion exchange and metabolic activity	pH 4.0, 25 °C, 10 g/L biomass concentration	73.96% at 20 g/L molasses concentration	Başkan and Açıkel (2021)
<i>Candida albicans</i>	Cr, Pb, Cu, Zn	Biosorption and intracellular accumulation	Cr(VI): pH 1, 60 °C, 24 h, 1 g/L biomass; Cu: pH 5, 28 °C	Cr(VI): 76%, Pb: 57%, Cu: 31%, Zn: 22%	Rodríguez et al. (2017)
<i>Candida krusei</i>	Cu	Biosorption via carboxylate chelation	pH 5.2, 40 °C, 24 h, 2 mL encapsulated biomass (MCaAlg)	Adsorption capacity: 0.687 mmol/g	Luk et al. (2017)
<i>Candida sp.</i>	Cu	Bioaccumulation; higher resistance to heavy metals compared to <i>S. cerevisiae</i>	pH 5–6, room temperature	Cu: ~4.8 mg/g	Das et al. (2008)
	Cu, Fe, Zn	Biosorption via cell wall functional groups	Effluent pH: 5.0, Temperature: 28 °C, Contact time: 120 min, Agitation: 150 rpm, Pre-treatments: NaOH	Cu: 97.6%, Fe: 98.6%, Zn: 96.7% (NaOH-treated biomass, 120 min). Untreated biomass: ~80% removal	Anaenene (2012)
	Pb, Cu	Biosorption through ion exchange and cell wall adsorption	pH 5–6, 28 °C, 12 h contact time	Pb: ~80%, Cu: ~85%	Wang and Chen (2009)
<i>Saccharomyces cerevisiae</i>	Cu, Zn, Cd	Biosorption via cell wall functional groups (e.g., carboxyl, phosphate)	pH 4–6, 25–30 °C, 24 h contact time	Cu: ~90%, Zn: ~75%, Cd: ~80%	



Table 2 (continued)

Microbial species	Heavy metal	Mechanism	Optimal conditions	Efficiency	Refs.
	Pb(II), Ni(II), Cr(VI)	Biosorption via carboxyl, hydroxyl, amino, and phosphate groups	pH: Pb(II) 5.0, Ni(II) 5.0, Cr(VI) 1.0. Temperature: 25 °C, Contact time: 24 h, Biomass: 1 g/L	Uptake: Pb(II) 270.3 mg/g, Ni(II) 46.3 mg/g, Cr(VI) 32.6 mg/g. Removal: Pb(II) > 95%, Ni(II) 82%, Cr(VI) 60%	Özer and Özer (2003)
	Cu, Zn, Cd, Pb	Biosorption via cell wall functional groups (e.g., carboxyl, phosphate); passive and active uptake	pH > 5 for Cd, Cu, and Zn, room temperature	Cu: ~ 7.11 mg/g (highest uptake among yeasts), Cd and Pb: moderate uptake	Das et al. (2008)
	Cd(II), Pb(II)	Adsorption via cell wall functional groups; Langmuir isotherm and pseudo-second-order kinetics	pH 4.0–5.5, 30 °C, contact time 30 min	Cd: 48.69% (0.126 mmol/g), Pb: 77.91% (0.214 mmol/g)	Dong et al. (2023)
	Cr(VI), Fe(III), Ni(II), Zn(II), Cu(II)	Adsorption via cell wall functional groups; fits Langmuir and Freundlich isotherms	pH 2 for Cr and Fe; pH 3–5 for Ni, Zn, Cu; 25–35 °C	Cr: 97.7%, Fe: 98.1%, Ni: 71%, Zn: 58%, Cu: 60%	Zinicovscaia et al. (2020)
	Cu(II)	Adsorption via functional groups on the cell wall; Langmuir isotherm model	pH 5, 30 °C, 24 h contact time, 2 g/L biomass concentration	34% efficiency at 25 mg/L Cu concentration	do Nascimento et al. (2019) and Anaemene (2012)
	U	Biosorption via reduction and adsorption	pH 5.0, 30 °C, 24 h contact time, 4.6 g/L biomass	51.5% in 360 min, sorption capacity: 27.0 mg/g	Chen et al. (2020)
	U, Zn, Cu	Adsorption on cell wall, intracellular	pH 4–5, 25 °C, 16 h contact time, dried cells	U: 0.592 mmol/g, Zn: 0.56 mmol/g, Cu: Dead cells: 0.16 mmol/g, Live cells: 0.2 mmol/g	Volesky and May-Phillips (1995)
	U	Ion exchange with carboxyl and phosphoryl groups	pH 4.0, 30 °C, contact time: 15 h, 5 g/L biomass	Adsorption capacity: 84.3% at pH 4.0	Zhang et al. (2020)



Table 3 Different types of filamentous fungi and heavy metal biosorption capacity

Division	Species	Heavy metal	Biosorption capacity	References
Glomeromycota	<i>Arbuscular mycorrhizal</i>	Pb	Not defined	Zhang et al. (2010)
Ascomycota	<i>Aspergillus niger</i>	Cu	Not defined	Luna et al. (2016)
		Cr	100%	Acosta-Rodríguez et al. (2018)
		As	66–98%	Duborská et al. (2020), Littera et al. (2011)
		Cd	57%	Kapoor et al. (1999)
		Pb	59%	Dursun (2006)
		Hg	83%	Kapoor et al. (1999)
	<i>Neurospora crassa</i>	Cr	Upto 95%	Tunali et al. (2005)
		Pb	Upto 95%	Sharma et al. (2021b)
	<i>Penicillium chrysogenum</i>	Pb	Pb: 96–116 mg/g	Deng and Ting (2005)
		Cd	21–56 mg/g	Skowroński et al. (2001)
		Cr	18–27 mg/g	Tan and Cheng (2003)
		As	33–57 mg/g	Loukidou et al. (2003)
	<i>Rhizopus arrhizus</i>	Pb	Pb: 200 mg/g	Fourest et al. (1994), Uslu et al. (2003)
		Cd	57.5–84%	Tobin et al. (1993)
		Cr	64–78 mg/g	Aksu and Balibek (2007)
	<i>Trichoderma viride</i>	Cd	Not defined	Singh et al. (2010), Morales-Barrera and Cristiani-Urbina (2006)
		Pb	Not defined	Ali and Hashem (2007)
		Cr	94–100%	Ali and Hashem (2007)
Basidiomycota	<i>Pleurotus ostreatus</i>	U	19.9 mg/g	Zhao et al. (2016)

metal-binding capabilities, and when present, these polysaccharides can be removed via chemical or mechanical methods (Shamim 2018). Typically, bacterial metal sequestration involves the cell wall's functional groups capturing metal cations, followed by their internalisation (Ansari et al. 2016; Priyadarshane and Das 2021).

The biosorption process initiates with metal ions interacting with the bacterial cell wall, binding to functional groups such as amine, carboxyl, hydroxyl, phosphate, sulphate, and amine (Abdi and Kazemi 2015; Vijayaraghavan and Yun 2008). This binding is succeeded by the internalisation of the ions, with Gram-positive bacteria exhibiting a higher uptake capacity due to glycoproteins. In contrast, Gram-negative bacteria show lesser uptake, attributed to their phospholipids and lipopolysaccharides content (Namdeti 2023; Shamim 2018).

Bioaccumulation further facilitates metal uptake in bacteria, where positively charged metals bind to negatively charged bacterial cell membranes, triggering polysaccharide release into the bacteria. Metals migrate from the outer cell membrane to the cytoplasm, ceasing movement in a process termed bioaccumulation (Ruiz-Fresneda et al. 2023; Binish et al. 2022). Immobilisation involves bacteria catalysing metal dissociation from minerals, predominantly executed by metal-reducing bacteria through the oxidation of organic compounds, producing reduced ferric ions (Sandhibigraha et al. 2020). This process includes the reduction of iron(III)

to iron(II), sulphate to hydrogen sulphide, and Mn(IV) to Mn(II), with the reduced products forming insoluble species. Ferrous ions, generated from iron-reducing bacteria with enzymatic capabilities, can precipitate and mitigate chromium and uranium in contaminated soils (Ramamamy et al. 2007). Sulphate-reducing bacteria play a crucial role in metal absorption, producing hydrogen sulphide capable of reducing and precipitating sulphide minerals such as pyrite (Acharya and Kharel 2020; Ramamamy et al. 2007). The metal biosorption capacities of various Gram-positive and Gram-negative bacterial species are detailed in Table 4.

The study of metal-binding proteins enriched with histidine residues reveals their function as natural reservoirs for heavy metals, contributing significantly to the redox balance within the cellular environment (Samuel et al. 2021). Proteins such as Hpn from *Helicobacter pylori* and SCO4226 from *Streptomyces coelicolor* have been identified as nickel-binding storage proteins, underscoring the diversity of heavy metal storage mechanisms beyond MTs within microorganisms (Chen et al. 2022; Hausinger 2022). This discovery suggests the existence of numerous yet-to-be-identified heavy metal storage systems within microbial entities (Diep et al. 2018).

Phytochelatin synthase is a key enzyme in bioaccumulating heavy metals within microorganisms. Genetic engineering to enhance the expression of genes such as *cysE*, *GshI*, and *GshII*— which are responsible for the biosynthesis



Table 4 Heavy Metal biosorption capacity of Gram-positive and Gram-negative bacteria

Type of bacteria	Species	Targeted heavy metal	Biosorption capacity	Reference
Gram + ve	<i>Bacillus xiamenensis</i>	Cd	99%	Charan et al. (2024)
		Cr	40%	
		Ni	85%	
	<i>Bacillus jeotgali</i>	Zn	128 mg/g	Green-Ruiz et al. (2008)
		Cd	53.5 mg/g	
	<i>Staphylococcus xylosus</i>	Cr	143 mg/g	Ziagova et al. (2007)
		Cd	250 mg/g	
	<i>Staphylococcus epidermidis</i>	Zn	59%	Quiton et al. (2018)
		Cr	56%	
	<i>Bacillus licheniformis</i>	Cu	94%	Patel and Chandel (2015)
		Fe	92%	
		Cr	52%	
		Cu	32%	
	<i>Micrococcus luteus</i>	Fe	95%	Puyen et al. (2012)
		Pb	36%	
		Cu	25%	
	<i>Arthrobacter sp.</i>	Pb	9.6 mg/g	Yu et al. (2016)
	<i>Streptomyces sp.</i>	Cd	68%	El-Motaleb et al. (2020) and Hamdan et al. (2021)
		Pb	83%	
		Cu	75%	
Gram – ve	<i>Pseudomonas sp.</i>	Cr	95 mg/g	Ziagova et al. (2007)
		Cd	278 mg/g	
		Fe	98 mg/Kg	
		Cu	96/Kg	
	<i>E. coli</i>	Ni	85.5%	Quintelas et al. (2009)
		Fe	100%	
	<i>Pseudomonas putida</i>	Zn	17.7 mg/g	Green-Ruiz et al. (2008)
	<i>Enterobacter sp. J1</i>	Cd	46.2 mg/g	Ziagova et al. (2007)
		Cu	32.5 mg/g	
	<i>Methylobacterium sp.</i>	Pb	62.3%	Jeong et al. (2019) and Photolo et al. (2021)
	<i>Pseudomonas aeruginosa</i>	Cr	93%	An et al. (2020)
	<i>Pseudomonas alcaliphila</i>	Cr	97%	El-Naggar et al. (2020)
	<i>Pseudomonas azotoformans</i>	Cd	98.5%	Choińska-Pulit et al. (2018)
		Cu	70%	
		Pb	78%	
	<i>Klebsiella sp.</i>	Hg	85%	Orji et al. (2021)

of cysteine, gamma-glutamylcysteine, and GSH, respectively—has been shown to increase the bioaccumulation capacity mediated by phytochelatin synthase significantly. For instance, the enzyme polyphosphate kinase in *Klebsiella pneumoniae* functions as an effective storage mechanism for various heavy metals, including arsenic, copper, and nickel, while showing particular efficiency in mercury bioaccumulation (Diep et al. 2018). Similarly, the bacterial species *Sphingomonas* produces alkaline phosphatase, an enzyme that catalyses the precipitation of uranyl phosphate from phosphorus ions, demonstrating its potential for uranium

accumulation (Kulkarni et al. 2013). These advancements illustrate how enzymatic pathways, when coupled with genetic engineering, can significantly enhance the capacity of microorganisms to bioaccumulate heavy metals, paving the way for effective bioremediation strategies in contaminated environments.

Beyond the synthesis of phytochelatin and specific enzymes, bioaccumulating microorganisms adjust their enzymatic responses upon heavy metal exposure by complexing with metals through the production of amino and organic acids. A notable instance is *Acidithiobacillus*



ferrooxidans, which modulates histidine synthesis in response to copper sulphate, increasing cytoplasmic histidine to chelate copper ions and mitigate oxidative stress (Diep et al. 2018). Different bacteria and their capacity to produce enzymes to bind metals are detailed in Table 5.

The induction of antioxidant enzymes in bacterial species subjected to heavy metal stress, utilising the minimum inhibitory concentration (MIC) of heavy metals as a quantitative measure, was investigated (Hussein and Joo 2013). The objective was to employ antioxidant enzymes as indicators for heavy metal accumulation within bacterial strains. The experimental methodology entailed isolating *Pseudomonas putida* strains from diverse soil samples, which were then exposed to heavy metal stress to measure antioxidant enzyme activity. This study demonstrated that heavy metals, particularly Zn^{2+} and Pb^{2+} , significantly increased antioxidant enzyme activity. For instance, Zn^{2+} exposure

at 100 mg/L resulted in a 1,030% increase in ascorbate peroxidase activity in *Pseudomonas putida* KNU5, while Pb^{2+} exposure induced lower levels of antioxidant enzyme activity (Hussein and Joo 2013). These findings suggest that heavy metal accumulation by bacteria may stimulate exponential growth phases under certain conditions. Yet, prolonged exposure leads to reduced bacterial counts, with the control sample showing the highest bacterial count at 2.1×10^7 CFU/g.

The MIC values were determined in liquid media for the most metal-resistant bacterial strains with elevated concentrations of heavy metals (Rathnayake et al. 2013). Interestingly, the MIC assessments on solid media exhibited higher values, likely attributable to more robust metal-binding interactions with the agar medium (Hussein and Joo 2013). For Pb and Zn, the MIC ranged between 600 and 700 mg/L in resistant bacterial strains, whereas, for

Table 5 Bioaccumulation of various metals and production of enzymes /metal binding proteins by bacterial species

Target heavy metal	Bacterial biosorbent	Bioaccumulation capacity	Initial conc. (mg/L)	Reference
Copper	<i>Escherichia coli</i>	145 mg/gDw	0.01	Ueki et al. (2003)
Cadmium	<i>Escherichia coli</i>	6 mg/gDw	56.1	Gong et al. (2015)
		10.5 mg/gDw	56.2	Deng et al. (2007)
	<i>Enterobacter cloacae</i>	71.2%	–	Ghosh et al. (2022)
	<i>Bacillus sp.</i>	76.4%	–	Heidari and Panico (2020)
	<i>Burkholderia cepacia</i>	> 90%	–	Zhang et al. (2019)
Nickel	<i>Deinococcus radiourans</i> R1	1.2%	0.0005	Gogada et al. (2015)
	<i>Escherichia coli</i>	0.88 mg/gDw	0.58	Krishnaswamy and Wilson David (2000)
		7.05 mg/gDw	75	Deng et al. (2005)
		1.2%	0.000002	Raghu et al. (2008)
		4.8 mg/gDw	0.17	Ben Ahmed et al. (2018)
Arsenic	<i>Escherichia coli</i>	0.17 mg/gDw	0.75	Kostal et al. (2004)
		0.61 mg/gDw	0.000035	Singh et al. (2011)
		0.32 mg/gDw	3.75	Su et al. (2009)
		5.24 mg/gDw	1000	Yang et al. (2014)
	<i>Corynebacterium glutamicum</i>	2.16 mg/gDw	7.5	Villadangos et al. (2014)
Mercury	<i>Escherichia coli</i> XLI-blue	1.07 mg/gDw	0.17	Pan-Hou et al. (2002)
	<i>Rhodopseudomonas palustris</i>	76 mg/gDw	90	Deng and Jia (2011)
	<i>Escherichia coli</i>	4.01 mg/gDw	6	Shahpiri and Mohammadzadeh (2018)
	<i>Vibrio fluvialis</i>	60%	–	Saranya et al. (2017)
	<i>Bacillus thuringiensis</i>	90%	–	Dash et al. (2014)
Uranium	<i>Escherichia coli</i>	151 mg/gDw	238	Basnakova et al. 1998)
	<i>Deinococcus radiodurans</i>	10,700 mg/gDw	2380	Kulkarni et al. (2013)
Multi metals	<i>Bacillus subtilis</i>	98% Pb, 93% Cd	–	Abioye et al. (2018)
	<i>Streptomyces zinciresistens</i>	98% Cd, 87% Zn	–	Lin et al. (2012)
	<i>Escherichia coli</i>	2.05 μ mol/gDw Cd, 1.81 μ mol/gDw Cu, 1.44 μ mol/gDw Zn	–	Li et al. (2021)
	<i>Pseudomonas sp.</i>	> 98% Pb, 75% Cd	–	De et al. (2008)



other metals, it spanned between 300 and 500 mg/L. Sensitivities varied, with MICs for Pb and Zn between 100 and 500 mg/L and other metals between 100 and 300 mg/L in strains vulnerable to metal toxicity. This variance underscores the substantial tolerance of certain bacterial strains to lead and zinc, potentially due to their adaptive capabilities in varying environmental contexts (Hussein and Joo 2013).

The interaction between heavy metals and bacterial strains was observed to generate reactive oxygen species, disrupting oxidative balance and prompting the synthesis of antioxidant enzymes such as CAT, Px, and ascorbate. *P. putida* exhibited a marked increase in CAT activity (311%) at a 100 mg/L Zn concentration. Contrastingly, the enzymatic induction of Px was comparatively lower than that of CAT and ascorbate peroxidase, with *B. subtilis* showing heightened Px activity at 100 mg/L of Pb and Zn. However, other strains, *P. putida* KNU8 and *P. putida* KNU5, exhibited reduced Px levels under similar conditions (Hussein and Joo 2013). This differential enzymatic response highlights the complex regulatory mechanisms bacteria employ in response to heavy metal exposure, with CAT production being predominant, followed by ascorbate peroxidase and then Px (Valencia-Marin et al. 2024; Lee 1997). In conclusion, this study demonstrates that bacterial species synthesise antioxidant enzymes as a defensive strategy against heavy metal toxicity. These enzymes offer a promising avenue for monitoring and understanding heavy metal bioaccumulation, both environmentally and within microbial systems (Yaashikaa et al. 2022; Khanna et al. 2018).

The interaction between soil microbiota and plant growth, particularly under heavy metal stress, underscores the pivotal role of arsenate-reducing bacteria in facilitating metal sequestration and enhancing plant biomass. A significant 44% increase in As accumulation was observed in the fern species *P. vittata*, accompanied by a 53% enhancement in overall plant biomass when in association with these bacteria (Yang et al. 2014). Further research highlighted the profound capability of endophytic bacteria to tolerate heavy metals, primarily due to their high metal-binding affinity, which not only boosts plant growth, including root development and spatial expansion, but also triggers the secretion of organic acids like oxalic, citric, and succinic by plants. These acids improve metal absorption mechanisms while alleviating plant oxidative stress (Ma et al. 2015).

It was shown that integrating various bacterial and fungal species, such as mushrooms, increases soil enzymatic activity, leading to higher metal accumulation levels than when these microorganisms are used in isolation (Li and Chen 2012). This synergy significantly contributes to the phytoremediation process, with multiple bacterial species investigated for their enhancement of plant metal absorption capabilities, as outlined in Table 6.

Using bacteria for metal accumulation augments plant growth and addresses the challenge posed by elevated heavy metal levels in soil, which can be toxic to plants. Microorganisms, especially bacteria, serve as effective biological agents in this context. By deploying specific enzymes and compounds to absorb heavy metals from the soil, these microorganisms facilitate changes in plant hormone levels, promoting unhindered plant growth. This microbial intervention reduces the adverse effects of heavy metals, thereby supporting sustainable plant development (Zhou et al. 2016). Such beneficial microorganisms are recognised as “plant growth-promoting bacteria”.

In an experimental study, the enzymatic activities of 1-aminocyclopropane-1-carboxylic acid deaminase (ACC deaminase) and indole acetic acid (IAA) production by plant growth-promoting bacteria were systematically monitored (Mendoza-Hernández et al. 2016). The study assessed ten bacterial strains from the rhizosphere of plants in mining waste environments, which exhibited at least three beneficial traits, including the production of IAA and siderophores, ACC deaminase enzyme activity, and the capability to solubilise inorganic phosphate. These strains were taxonomically identified as belonging to the *Enterobacter*, *Serratia*, *Klebsiella*, and *Escherichia* genera. The activities of ACC deaminase and IAA synthesis were quantified under heavy metal exposure, with Pb, As, and Cu notably enhancing both ACC deaminase activity and IAA synthesis in specific bacterial strains, contrasting with the adverse effects of Ni, Cd, and Mn on these processes. Notably, the *Serratia* K120 strain demonstrated enhanced ACC deaminase activity and IAA production under heavy metal stress, promoting growth in *Helianthus annuus* plants and suggesting its potential in phytoremediation strategies (Mendoza-Hernández et al. 2016).

This research underscores the strategic use of PGPB, including recognised strains such as *Pseudomonas*, *Klebsiella*, and *Enterobacter*, to enhance plant resilience to heavy metals through bioaccumulation. Studies have demonstrated that these bacteria, particularly under exposed to metals like Cu, Pb, and As, produce IAA as a protective response, facilitating plant growth (Ahmad et al. 2016; Kamnev et al. 2005). Additionally, the role of siderophores in augmenting IAA production under heavy metal exposure emphasises the complex interactions between microbial agents and plant responses to environmental stressors, offering promising avenues for bioremediation and sustainable agriculture (Ahmad et al. 2016; Kamnev et al. 2005).

Performance indicators involved in the biosorption of heavy metals

Biosorption has emerged as an effective, cost-efficient, and environmentally sustainable strategy for mitigating heavy metal pollution, a consequence of widespread industrial



Table 6 Bacterial mechanisms for enhancing metal absorption by plants

Bacterial Strain	Heavy Metal	Mechanism	Optimal Conditions	Efficiency	Refs.
<i>Pseudomonas korensis</i> AGB-1	Cd, Pb, Cu, Zn, As	Enhances plant biomass and metal uptake through IAA production and ACC deaminase activity	Soil pH: 5.6, Temperature: 28–32 °C, Soil water content: 70% w/w, natural light	Increased biomass (54%), chlorophyll (27%), and protein (28%). Root HM uptake: As (23%), Cd (31%), Cu (7%), Pb (18%), Zn (15%)	Babu et al. (2015)
<i>Pseudomonas</i> sp. Lk9	Cd, Zn, Cu	Biosurfactant and siderophore production; organic acid secretion; enhanced nutrient availability	Soil pH: 7.8, Temperature: 25 °C, 16/8 h light cycle	Cd: 46.6% (shoot), Zn: 16.4%, Cu: 16.0%	Chen et al. (2014)
<i>Bacillus thuringiensis</i> GDB-1	Ni, Cd, Zn, Pb, Cu, As	IAA production, ACC deaminase activity, siderophore production, phosphate solubilisation	Soil pH: 6.5, Temperature: 28–32 °C, natural light	Pb: 77%, Zn: 64%, As: 34%, Cd: 9%, Cu: 8%, Ni: 8%	Babu et al. (2013)
<i>Bacillus</i> sp. SLS18	Cd, Mn	Produces IAA, siderophores, and ACC deaminase; enhances biomass and metal uptake	Soil pH: 6.65, Temperature: 28/25 °C (day/night), 14-h light	Mn uptake: 65.2%, Cd uptake: 40.0% (aerial parts, <i>Sorghum bicolor</i>)	Luo et al. (2012)
<i>Bacillus pumilus</i> E2S2	Cd, Zn	Produces IAA, ACC deaminase, and siderophores; enhances biomass and metal uptake	Soil pH: 7.6, Temperature: 28 °C, natural light	Cd: 43% (plant uptake), Zn: 18% (plant uptake),	Ma et al. (2015)
<i>Bacillus</i> sp.	As(V), As(III)	Arsenate (As(V)) reduction and arsenite (As(III)) oxidation; limited IAA and siderophore production	10 mM As(V) and As(III), pH neutral, 30 °C	As(V) reduction: ~26% (strain PML02); As(III) oxidation: ~36% (strain PVS06)	Zhu et al. (2014)
<i>Rahnella</i> sp. JN6	Cd, Pb, Zn	Produces IAA, siderophores, ACC deaminase; enhances metal solubilisation	Soil pH: 6.65, Temperature: 28 °C, 16/8 h light cycle	Cd: 49% increase (shoot), Pb: 47% increase, Zn: 28% increase in uptake	He et al. (2013)
<i>Rahnella</i> sp. JN27	Cd	Produces IAA, siderophores, ACC deaminase; enhances Cd solubilisation	Soil pH: neutral, Temperature: 28 °C, light regime: 14/10 h day/night	Increased Cd uptake in shoots: 144.4% (<i>A. hypochondriacus</i>); 100.5% (<i>A. mangostanus</i>) at 100 mg Cd/kg soil	Yuan et al. (2014)
<i>Exiguobacterium</i> sp. As-9	As(V), As(III)	Arsenate reductase for As(V) reduction; arsenite oxidase for As(III) oxidation; intracellular accumulation	pH 8, 37 °C, aerobic, 10-day incubation	As(V): 99.5% removal, As(III): 85% removal	Pandey and Bhatt (2016)



activities. This method capitalises on the intrinsic affinity of biomass for heavy metals, employing mechanisms such as ion exchange, surface complexation, and microprecipitation to facilitate metal removal (Torres 2020; Giese 2020; Dinh et al. 2021). Key to this process are functional groups—including carboxyl, hydroxyl, sulphate, phosphoryl, and amino groups—integral to the metal binding efficiency. Several factors can influence the effectiveness of metal absorption via biosorption, including the selectivity of metal ions, modifications to the biosorbent, and the overall capacity of the biosorbent (Luka et al. 2024). Additionally, various physicochemical parameters, such as temperature, solution pH, contact time, agitation speed, and co-ion presence, significantly impact biosorption (Luka et al. 2021; Hussein and Joo 2013; Abo-Alkasem et al. 2023; Sabreena et al. 2023).

Biosorbent quantity

Due to the specific mechanisms involved, the biosorbent and biosorbate's nature regulates biosorption processes, particularly in chemisorption. In contrast, physisorption is independent of the biosorbent type. Each biosorbent has unique features and functional groups that serve as primary metal binding factors (Luka et al. 2024). It was reported that increasing the biosorbent concentration reduces the amount of biosorbed biosorbate per unit weight of biosorbent while improving its overall removal rate. It was observed that for bacterial cells with biosorbate biosorption at an optimal pH, an increased biosorbent quantity lowers the maximal specific biosorbate uptake, likely due to cell aggregation. At acidic pH levels, this trend reverses, possibly due to partial breakdown of the bacterial cell wall. The dependency on biosorbent concentration can be attributed to electrostatic interactions between cells, interference between binding sites, and reduced mixing (Hussein and Joo 2013).

The biosorbent quantity significantly influences the biosorption of heavy metals, as it provides the sorption sites for metal ions. A higher biosorbent dose at a given initial metal concentration increases the percentage of biosorption due to the larger surface area, which enhances the number of available sorption sites but decreases the metal uptake per unit mass of biosorbent. Thus, comparing sorbents using percentage biosorption is crude and inaccurate, serving only for quick and approximate screening of biosorbents. A more accurate approach is to use metal uptake per unit mass of adsorbent (Olawale 2020). An increase in the percentage of Cr biosorption was reported using *Parthenium hysterophorus* from 61.28 to 80.81%, with an increase in biosorbent dosage from 0.1 to 1 g due to the availability of more sorption sites, but a decrease in metal uptake from 9.43 to 0.37 mg/g because of the increased biosorbent-to-metal ratio (Venugopal and Mohanty 2011).

Temperature

Temperature further affects biosorption dynamics, with higher temperatures potentially enhancing the biosorption rate by increasing surface activity and kinetic energy. However, the impact on the maximum capacity for metal biosorption is variable. An increase in temperature typically favours endothermic processes, possibly due to structural alterations in the biosorbent or the dissolution of intermolecular bonds. Notably, the response of living biomass to temperature fluctuations differs from that of dead biomass, exhibiting increased metabolic activity at elevated temperatures up to a certain threshold, beyond which biosorption efficiency declines (Gouda and Taha 2023; Libatique et al. 2020; Shamim 2018; Moreira et al. 2019).

Solution pH

The pH of the environment is the most critical factor influencing the biosorption process, with a neutral to slightly alkaline range (pH 7.0–8.0), optimising the attraction of cationic metal species due to the biosorbent's negative charge. It affects the solubility of the biosorbate and the activity of functional groups in biosorbents. For instance, hydrogen and hydroxide ions are strongly adsorbed, while the adsorption of other ions is dependent on the solution's pH (Luka et al. 2024). In contrast, metals that are predominantly anionic, such as chromium, arsenic, and molybdenum, are more efficiently captured at acidic pH levels (pH 2.0–4.0) (Namdeti 2023; Sedlakova-Kadukova et al. 2019; Shamim 2018; Torres 2020).

Contact time

Contact time with the aqueous solution is a crucial parameter for a successful biosorption process. Metal ion biosorption generally involves two stages: an initial rapid uptake phase, where approximately 90% uptake occurs within an hour, and a slower phase, where equilibrium is achieved. The rapid initial uptake is due to the availability of a large surface area for metal sorption. As time progresses, the biosorption rate decreases due to the depletion of surface biosorption sites. Fast primary sorption is attributed to extracellular binding, while slower sorption is due to intracellular binding (Luka et al. 2024; Olawale 2020).

Nature of biomass

A wide variety of biomaterials have been utilised for the biosorption of metal ions. Studies indicate dead biomass is more effective for metal ion sorption than living biomass. Chemical pretreatment of biomass, such as alkali treatment, enhances biosorption efficiency. Physical treatments such as



boiling, crushing, drying, and mechanical disruption also impact biosorption performance (Olawale 2020).

The deacetylation of fungal cells alters the structure of chitin, resulting in the formation of chitosan-glycan complexes with high metal affinity (Wang and Chen 2006). All factors, such as age and the growth media components, influence biosorption by affecting cell wall composition, cell size, and the production of extracellular polymeric substances (Namdeti 2023).

Initial metal ion concentration

The initial metal ion concentration is a crucial driving force to overcome mass transfer resistance between the solid and aqueous phases (Olawale 2020). The metal uptake (mass of sorbed metal per unit mass of adsorbent) of the biosorbent initially increases with rising metal ion concentration until saturation is reached. However, high metal ion concentrations decrease removal efficiency due to reduced interaction of the ions with available sorption sites (Luka et al. 2024; Namdeti 2023; Olawale 2020; Shamim 2018). For instance, increasing the initial metal ion concentration from 20 to 120 mg/L significantly enhanced the uptake of Cr, Cu, and Pb by termite feathers, with a slight increase for Cd (Olawale 2017).

Presence of Co-ions

The coexistence of other sorbable cations with the target metal leads to competition for active binding sites or the formation of complexes, consequently inhibiting or reducing the biosorptive removal of the target metal ion. The degree of this inhibition or reduction is contingent on the binding affinities of the involved cations. Studies have shown that magnesium and sodium significantly compete with zinc for binding sites on algae, resulting in approximately a 90% reduction in zinc biosorption at 200 mM NaCl or $\text{Mg}(\text{NO}_3)_2$ concentrations. Conversely, the interference of these cations with the biosorption of lead or copper is much less pronounced, with less than a 50% reduction in the sorption of lead or copper at the same concentration levels of NaCl or $\text{Mg}(\text{NO}_3)_2$ (Olawale 2020).

Despite this challenge, biomass—particularly from algae and microorganisms—remains distinguished by its high adsorption efficiency and the variety of functional groups available for metal interaction. Exploring genetic modifications to microbial biomass to improve metal biosorption capacities, such as expressing specific proteins or MTs, has attracted considerable research, demonstrating substantial increases in biosorption capabilities for various metals (De Oliveira et al. 2020; Dong et al. 2019; Lin et al. 2020; Tan et al. 2019; Torres 2020).

Moreover, the significance of biosorption extends beyond metal removal to include metal recovery, offering an economic benefit to this remediation technique. For biosorption to be viable at an industrial scale, the adaptability and customisation of biomass to meet commercial demands are essential. This adaptability and the prospect of metal recovery position biosorption as a promising and globally relevant approach to addressing metal pollution challenges.

Agitation speed

Increasing the agitation speed enhances the biosorption rate of the sorbate by reducing its mass transfer resistance. However, excessively high agitation speeds can damage the structure of fragile sorbents. Optimal agitation speed ensures homogeneity and maximises contact between the sorbate and sorbent. At low agitation speeds, insufficient contact occurs, while moderate speeds provide the best homogeneity and effective biosorption. Conversely, high agitation speeds can create a vortex, leading to a loss of homogeneity and reduced biosorption performance due to decreased contact time between the biosorbent and metal ion (Gouda and Taha 2023; Luka et al. 2024; Olawale 2020).

Studies have identified an optimal agitation speed of 120 rpm for the biosorption of cadmium and zinc using *Aspergillus niger*. Additionally, the biosorption efficiency increased significantly from 32.4 to 65% when the agitation speed was raised from 0 to 80 rpm (Liu et al. 2006; Yeddou-Mezenner 2010).

Biosorption mechanisms

Biosorption of heavy metals by biological materials occurs through various mechanisms, each influenced by the chemical properties of the metal ion and the nature of the biosorbent. The primary mechanisms include physical adsorption, ion exchange, complexation, and precipitation. Understanding these mechanisms is crucial as each has distinct advantages depending on environmental conditions, such as pH, temperature, and the specific metal involved. This section will compare these mechanisms to elucidate under which conditions one might be preferred over another, providing a comprehensive understanding of practical applications (Wenten et al. 2020).

Physical adsorption

Physical adsorption involves the attraction of metal ions to the biosorbent surface through van der Waals forces or electrostatic interactions. This mechanism is typically favoured in scenarios where the biosorbent has a large surface area, allowing for greater contact between the metal ions and the



adsorbent. The effectiveness of physical adsorption is highly dependent on the pH of the solution, which influences the charge on the biosorbent surface and the metal ions. For instance, chromium(III) ions are more effectively adsorbed through physical adsorption when using biosorbents like grass or wheat straw in neutral pH conditions, where a monolayer adsorption process is dominant (Tripathi et al. 2023; Poonam et al. 2021). The simplicity and reversibility of physical adsorption make it an attractive option for rapid metal removal, particularly in the initial treatment stages.

Ion exchange

Ion exchange involves the replacement of metal ions in the solution with ions from the biosorbent. This mechanism is particularly effective when the biosorbent contains polysaccharides with functional groups, such as carboxyl and hydroxyl groups, that can readily swap ions with metal cations in the solution. The process is highly pH-dependent; for example, ion exchange between heavy metals like lead and cadmium is enhanced at lower pH levels with cations in the alginate of brown algae. However, at higher pH levels, other mechanisms, such as precipitation, may dominate (Ali Redha 2020). Ion exchange is advantageous when high selectivity and recovery of specific metals are required, making it suitable for applications where precise metal removal is critical.

Complexation

Complexation involves the formation of stable complexes between metal ions and functional groups present in the biosorbent, such as carboxyl, amino, or sulfhydryl groups. The efficiency of complexation is influenced by the metal ion's affinity for these groups, which can be predicted by the hard and soft acids and bases theory. For example, soft acids like lead strongly prefer complexation with soft bases such as sulfur-containing ligands in biosorbents (Abo-Alkasem et al. 2023). Complexation is particularly useful for the selective removal of metals that form stable complexes, providing a mechanism that can be fine-tuned based on the specific metal ions targeted for removal.

Precipitation

Precipitation occurs when metal ions react with components of the biosorbent or the surrounding environment to form insoluble compounds, precipitating out of the solution. This mechanism will likely happen in higher pH environments, where metal hydroxides become insoluble and form precipitates. Precipitation is significant in removing metals like iron and chromium, which can precipitate from solution at neutral to slightly basic pH levels, thereby reducing their

bioavailability (Brazesh et al. 2021). While effective, precipitation may lead to challenges in handling solid residues, necessitating additional steps for sludge management.

Practical implications

In practice, the choice of biosorption mechanism is influenced by the specific environmental conditions and the desired outcome of the treatment process. Physical adsorption may be preferable in situations requiring rapid metal removal, whereas ion exchange and complexation might be more effective in achieving high selectivity and recovery of specific metals. Precipitation, while effective under certain conditions, may lead to challenges in handling solid residues. Therefore, a thorough understanding of these mechanisms and the conditions that favour them is essential for optimising biosorption processes in real-world applications (Wenten et al. 2020).

Techniques for enhanced metal absorption by microorganisms

Genetic engineering

Recent studies highlight the enhanced capabilities of genetically modified microorganisms in biosorption. These modifications increase the adsorption capacity and specificity for target metals, making the process more efficient (Fulke et al. 2020; Tripathi et al. 2023). Applying genetic engineering to bacterial, fungal, and algal species represents a burgeoning strategy to amplify heavy metal accumulation from waste. Enhancing enzymatic activities by introducing exogenous genes promises improved metal absorption and efficacious bioremediation (Pant et al. 2021; Saravanan et al. 2023, 2022). A notable example includes a genetically modified *Escherichia coli* strain endowed with the MT gene, which showcased a Cd biosorption efficiency exceeding 75% within the first 20 min of exposure (Gadd 2009). Furthermore, phytochelatins, with their γ Glu-Cys repeating units, exhibit enhanced metal-binding capabilities, particularly for Cd and Hg ions, underscoring the potential of genetic augmentation of the metal-binding motif (Glu-Cys) in biosorption processes (Bae et al. 2000).

In the realm of genetic modifications, Gram-positive bacteria, characterised by a singular peptidoglycan layer, are preferred due to their robustness against shear forces. Engineered strains of *Staphylococcus xylosus* and *Staphylococcus carnosus*, expressing chimeric polyhistidyl peptides, have demonstrated significant improvements in Ni binding, facilitating Ni removal from industrial effluents (Patel and Chandel 2015). Moreover, genetically engineered *E. coli* strains subjected to Hg showcased notable mercury absorption,



particularly at lower concentrations, highlighting the efficacy of genetic engineering in enhancing metal uptake (Diep et al. 2018). This augmentation in uptake capacity extends across various microorganisms, where genetically modified organisms absorbed higher concentrations of heavy metals compared to their wild-type counterparts (Mishra and Malik 2013). These findings underscore that genetic engineering not only amplifies the inherent capabilities of microorganisms but also enables them to bioaccumulate metals beyond the natural growth inhibition limits of wild strains (Ojha et al. 2023; Saif et al. 2021).

Building on these promising outcomes, further applications of genetic engineering have broadened the scope of microbial heavy metal remediation. For example, *Alcaligenes eutrophus* AE104 (pEBZ141) has been utilised for chromium removal from industrial wastewater (Srivastava et al. 2010), and a recombinant photosynthetic bacterium, *Rhodospseudomonas palustris*, was engineered to simultaneously express a mercury transport system and metallothionein, enabling efficient Hg^{2+} removal from heavy metal wastewater (Deng and Jia 2011). Chromosomally located PCB catabolic genes from *R. eutropha* A5, *Achromobacter* sp. LBS1C1 and *A. denitrificans* JB1 were transferred into a heavy metal-resistant strain, *R. eutropha* CH34, through natural conjugation for the degradation of polychlorinated biphenyls (PCBs) (Springael et al. 1994).

Genetic engineering of endophytic and rhizospheric bacteria for plant-associated degradation of toxic compounds in soil is emerging as one of the most promising technologies for environmental remediation (Bhatia and Kumar 2011). To select an appropriate strain for gene recombination and inoculation into the rhizosphere, three criteria have been recommended: (1) the strain should be genetically stable after cloning, with high expression of the target gene; (2) it should be tolerant or insensitive to the contaminant; and (3) it must be able to survive in specific plant rhizospheres (Doukani et al. 2023). Many rhizospheric bacteria exhibit limited abilities to degrade organic pollutants. However, advances in molecular biology have enabled the development of genetically engineered rhizobacteria carrying contaminant-degrading genes, significantly enhancing their effectiveness in rhizoremediation. These engineered bacteria hold promise for the remediation of diverse environmental contaminants, contributing to sustainable soil and water management in polluted sites (Glick 2010).

Building upon these advancements, incorporating specific promoters and binding sites into microorganisms allows for the regulation of import storage systems upon heavy metal exposure. This targeted approach to bioaccumulation enhances the capacity of genetically modified microorganisms, characterised by a meticulously analysed import system comprising proteins optimised for heavy metal accumulation. By inducing the production of metal-binding proteins

and enzymes, genetic engineering increases the ability of bacteria to sequester heavy metals and mitigates the toxicity arising from oxidative stress, offering a promising avenue for the bioremediation of contaminated environments (Diep et al. 2018).

Genetic engineering of microalgae also presents a novel approach for recovering precious metals, including gold. This technique enables the selective absorption of gold by microalgae, facilitating its subsequent extraction from the cells (Ummalyma and Anamika 2021). *Chlorella vulgaris*, a type of microalgae, demonstrates significant potential for metal ion biosorption (Darnall et al. 1986). The research explored two methods: direct interaction and immobilisation of the algae to assess the adsorption of Au, Hg, and Ag ions. Results revealed that while Cr, Fe, Al, Zn, and Ni strongly bind to *Chlorella* at pH 5–7, Hg, Ag, and Au required pH adjustments for optimal binding. Notably, Au showed the highest binding efficiency, with 90% recoverable from dilute solutions, highlighting *Chlorella vulgaris* as a promising tool for metal recovery in mining operations.

Following biofuel extraction, a study explored microalgae residue as an adsorbent for recovering metal ions, particularly gold (Kanjana 2008). The microalgae were pre-treated with sulfuric acid and sodium bicarbonate before being pulverised into cross-linked particles. In acidic chloride solutions, these cross-linked algae exhibited a strong affinity for metal ions, with gold showing the highest adsorption, followed by palladium and platinum. The cross-linking process significantly enhanced the adsorption capacity for gold, achieving 3.25 mol/kg. Microscopic analysis further confirmed the formation of gold particles within the algae, underscoring the potential of microalgae-derived materials for gold recovery applications.

While the application of genetic engineering and nanotechnology in biosorption offers tremendous potential, it also raises risks and ethical concerns that must be carefully considered. One of the primary risks associated with genetic engineering is the unintended ecological impact that genetically modified organisms might have if released into the environment. These organisms could outcompete native species, disrupt local ecosystems, or transfer their engineered traits to wild populations, leading to unforeseen consequences (Bae et al. 2000; Saif et al. 2021).

Similarly, using nanotechnology in biosorption introduces concerns about the toxicity and long-term environmental effects of nanoparticles (NPs). While NPs can enhance the efficiency of biosorption processes, their small size and high reactivity pose challenges in predicting their behaviour in natural environments. There is a risk that NPs could accumulate in the food chain, leading to potential health hazards for wildlife and humans (Gadd 2009; Saravanan et al. 2023).

Ethically, deploying genetically modified organisms and nanotechnology in environmental applications requires a



careful balance between innovation and precaution. Regulatory frameworks must be developed to ensure these technologies are used responsibly, with thorough risk assessments and ongoing monitoring to mitigate potential negative impacts. Public engagement and transparency in decision-making processes are critical to addressing societal concerns and fostering trust in these emerging technologies (Darnall et al. 1986; Saif et al. 2021).

Nanotechnology

Nanotechnology has recently introduced innovative environmental treatment methods, particularly for removing toxic pollutants like heavy metals and dyes. Due to their distinct properties, NPs are adaptable and versatile support systems for living cells (Azeez et al. 2022). Nanotechnology plays its role in the bioremediation of heavy metals in two ways: (i) nano-bioremediation and (ii) nanocell immobilisation.

Nano-bioremediation is an emerging technology that uses NPs produced by bacteria, algae, fungi, actinomycetes, and plants to remove environmental pollutants from contaminated sites. Leveraging nanotechnology, this approach offers an eco-friendly and cost-effective solution for environmental cleanup, providing a promising alternative for eliminating contaminants from the environment (Ramezani et al. 2021). Nano-bioremediation was defined as the process of using NPs along with microorganisms or plants to remove contaminants (Koul and Taak 2018). This concept was then refined by categorising the practices based on the type of organism used, introducing terms like phyto-nanoremediation, microbial nanoremediation, and zoo-nano-remediation (El-Ramady et al. 2017). Effective bioremediation relies on the successful interaction between NPs and living organisms. However, factors such as nano-toxicity, the size of the NPs, and nano-nutrition can impact these organisms, potentially influencing the overall effectiveness of the bioremediation process (Vázquez-Núñez et al. 2020). Bioremediation using nanomaterials has emerged as an effective strategy for addressing heavy metal contamination, particularly with mercury (Hg), cadmium (Cd), and lead (Pb). These metals are highly toxic and pose significant environmental and health risks due to their persistence and bioaccumulative properties. Nanomaterials, like zero-valent iron NPs and carbon nanotubes, can transform toxic mercury into less harmful forms, reducing its environmental impact. For cadmium, nanomaterials such as iron oxide NPs and zeolites effectively adsorb and immobilise the metal, preventing migration. In the case of lead, nanomaterials like hydroxyapatite NPs and modified clay minerals can adsorb lead ions and stabilise them, thereby reducing their bioavailability and preventing groundwater contamination (Kaur and Roy 2021; Nandini et al. 2023).

Applying cell immobilisation techniques to NPs has garnered significant interest from researchers, particularly for scaling these methods to an industrial level. Immobilising microbial cells enhances their stability and resilience to environmental conditions (Azeez et al. 2022).

Arsenic contamination in water, a global issue, is effectively addressed through biosorption using microbial biomass. *Arthrobacter* sp. can absorb 75 mg of As(III) and 82 mg of As(V) per gram of cells (Prasad et al. 2013). *Corynebacterium glutamicum* immobilised on neem leaves/MnFe₂O₄ and sawdust/MnFe₂O₄ composites demonstrates high adsorption capacities, particularly in the presence of Cd(II) and Fe(III) ions. These biosorbents, with maximum capacities up to 1861.71 mg/g for As(V), show potential for large-scale arsenic removal in water treatment systems (Podder and Majumder 2016).

The removal efficiency of Cu(II) was reported to be 96.8% when *Saccharomyces cerevisiae* was immobilised on chitosan-coated magnetic NPs under optimal conditions (Peng et al. 2010). Similarly, it was found that immobilising *Aspergillus niger* spores in Ca-alginate shield them from environmental factors, maintaining their high biosorption efficiency for Cr(VI) (Ren et al. 2017). Cadmium, a highly toxic metal, has been effectively adsorbed using chitosan-based biosorbents with reactive groups like –OH and –NH₂ (Madala et al. 2017). In a study, *Kluyveromyces fragilis* and magnetic chitosan beads efficiently removed Sr-90, reaching equilibrium swiftly (Ji et al. 2010). Algal biomass such as brown algae (*Sargassum glaucescens*) and red algae (*Gracilaria corticata*), and nanoscale zero-valent iron NPs have also been employed for heavy metal removal due to their cost-effectiveness and unique properties (Giese 2020; Giese et al. 2020). Furthermore, *Aspergillus ficuum* and *Pseudomonas aeruginosa* were studied for extracting thorium, uranium, and rare earth elements, while magnetically modified *Rhodotorula glutinis* and *S. cerevisiae* cells showed effective biosorption of U(VI), highlighting their potential for rapid and efficient heavy metal remediation (Desouky et al. 2016; Bai et al. 2012).

Future perspective of biosorption

It is challenging to pinpoint a single microorganism or method as the best for absorbing heavy metals from polluted environments, as each approach has its strengths. Some researchers advocate using fungal biomass for heavy metal biosorption from wastewater, while others see algae as the future for sustainable biosorption, especially in treating industrial effluents (Taha et al. 2023; Ordóñez et al. 2023). Life cycle assessments and techno-economic analyses indicate that immobilised algae are particularly effective for heavy metal removal (Chen et al. 2023). Hybrid technologies



that combine microalgae and bacteria could significantly improve the efficiency and performance of large-scale wastewater treatment. The symbiosis between microalgae and fungi is promising, as it reduces culture costs, increases biomass, and simplifies the solid–liquid separation process. It was found that co-immobilising microalgae and fungi with a mycelium-particle flocculant enhanced Cd(II) resistance and achieved a high adsorption efficiency of 98.89% (Wang et al. 2021). However, most research on immobilised algae remains at the laboratory stage, with limited progress toward large-scale industrial applications, and lacks specific life cycle assessments.

Additionally, processes like desorption, regeneration, and reuse of biosorbents need further pilot-scale testing. Algae have been highlighted for their potential in biosorption due to their extensive surface area, natural abundance, and capacity to accumulate heavy metals and other pollutants (Staszak and Regel-Rosocka 2024). Various environmental factors, such as pH, temperature, and biomass concentration, are critical for effective large-scale biosorption. Combining emerging technologies like immobilisation, genetic engineering, nanotechnology, and artificial intelligence is expected to enhance future research efforts, providing more sustainable solutions for heavy metal removal.

Conclusion

The increasing contamination of ecosystems with heavy metals presents a significant environmental and public health challenge, necessitating the development of sustainable remediation strategies. Microbial biosorption has emerged as a viable and eco-friendly alternative, leveraging the natural metal-binding capabilities of microorganisms such as bacteria, fungi, and algae. This review has highlighted the multifaceted interactions between these microorganisms and heavy metals, elucidating the mechanisms underpinning biosorption and their implications for environmental restoration.

Microbial biosorption offers several advantages over traditional remediation methods, including lower operational costs, reduced secondary pollution, and the ability to target a broad spectrum of metal ions. The adaptability of microbial biomass to various environmental conditions further enhances its potential as a sustainable remediation tool. The classification and characteristics of different biosorbent microorganisms reveal a vast potential for biosorption applications across diverse contaminated environments.

Recent advancements in genetic engineering and biotechnological innovations have further enhanced the biosorptive efficiency of microorganisms, paving the way for more targeted and efficient bioremediation strategies. The ability to genetically modify microorganisms to improve their

metal-binding properties and adapt them to specific environmental conditions represents a promising avenue for future research.

Despite the significant progress made, several challenges remain, including the need for large-scale application studies, the development of efficient biosorbent recovery and reuse methods, and the integration of biosorption with other remediation technologies. Future research should address these challenges and explore the synergistic effects of combining microbial biosorption with other emerging technologies, such as nanotechnology and artificial intelligence, to develop more robust and effective remediation strategies.

As the urgency to address heavy metal pollution intensifies, the practical applications of microbial biosorption in industries such as mining, wastewater treatment, and agriculture become increasingly relevant. Integrating microbial biosorption with cutting-edge technologies offers a pathway to innovative solutions that can mitigate heavy metal pollution, protect environmental and public health, and contribute to a more sustainable future.

In conclusion, microbial biosorption represents a cornerstone in the quest for sustainable and effective environmental remediation. By harnessing the natural capabilities of microorganisms and integrating them with cutting-edge technologies, we can develop innovative solutions to mitigate heavy metal pollution and protect environmental and public health for future generations.

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Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.

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